

LIBRARY DATABASE SYSTEM

Database Management System Presentation

Platform
MySQL 8.0.16 or newer

Core data
Books, members, loans, reservations

Automation
Triggers, procedures, function and view

Based on the Library Database Beginner's Guide

1. System Overview

What the database manages

- Stores organised information for a small library.
- Records members borrowing books through the loans table.
- Supports reservations when all copies of a book are borrowed.
- Creates fines automatically when books are returned late.
- Records fine payments and important changes in an audit log.
- Uses triggers for automatic rules and stored procedures for user-invoked operations.

2. Key Database Concepts

Table
Rows and columns

Primary Key
Unique record ID

Foreign Key
Links tables

Constraint
Protects valid data

Trigger
Runs automatically

Stored Procedure
Saved steps using CALL

Function
Saved calculation

View
Saved query

Index
Faster searching

3. The 9 Tables

Main entities in the library system

- categories — subjects such as Law or Literature
- authors — author names
- books — title, ISBN and copies
- members — name, email, phone and status
- loans — borrowing, due and return dates
- reservations — waiting-list records
- fines — late fees
- payments — money paid toward fines
- audit_log — history of important changes

4. Relationships Between Tables

Foreign keys connect the library data

Authors → Books
One author can have many books.

Categories → Books
A category can contain many books.

Members → Loans
A member can have multiple loans.

Books → Loans
A book can appear in many loans.

Loans → Fines
Each fine points to a loan.

Fines → Payments
Payments link to a fine.

Audit Log
Records important changes.

5. Constraints and Data Integrity

- PRIMARY KEY gives every row a unique identifier.
- AUTO_INCREMENT generates the next identifier.
- NOT NULL prevents required values from being empty.
- UNIQUE prevents duplicate emails or ISBNs.
- DEFAULT supplies a value when none is given.
- CHECK validates values against a condition.
- ENUM restricts values to an allowed list.
- FOREIGN KEY ensures relationships point to real records.

6. Stored Function and Procedures

`fn_calculate_fine`

Calculates late fines. The guide us

`sp_add_member`

Adds a new member.

`sp_available_books`

Lists books with a free copy.

`sp_member_open_loans`

Shows one member's books still c

`sp_return_book`

Returns a book; triggers handle th

```
SELECT fn_calculate_fine('2026-10-01',  
-- 3 days late → 1500.00 UGX
```

7. Triggers: Automatic Helpers

Triggers respond to changes in table data

- BEFORE INSERT — runs before a row is saved and can stop or change it.
- AFTER INSERT — runs after a row is saved.
- AFTER UPDATE — runs after a row changes.
- BEFORE DELETE — runs before deletion and can stop it.
- Examples: block invalid loans, update copies, create fines, update members

```
DELIMITER //  
CREATE TRIGGER trg_members_bi  
BEFORE INSERT ON members  
FOR EACH ROW  
BEGIN  
    SET NEW.email = LOWER(TRIM(NEW.email));  
END //  
DELIMITER ;
```


8. Four New Simple Triggers

trg_members_bi

Cleans names and converts emails to lowercase.

trg_books_bi

Starts a new book with all copies available.

trg_loans_ai_log

Writes a BORROW entry to audit_log.

trg_payments_ai_log

Writes a PAYMENT entry to audit_log.

```
SET NEW.full_name = TRIM(NEW.full_name);  
SET NEW.email = LOWER(TRIM(NEW.email));
```

9. Borrowing and Returning Workflow

1. Borrow

Checks member status, 3-book limit

2. Save Loan

Loan is saved when checks pass

3. Update Stock

Available copies decrease.

4. Return

CALL `sp_return_book(loan_id)`.

5. Waiting List

Oldest reservation becomes ready

6. Fine

Late return creates a fine at 500 UGX

10. Reservations and Waiting List

- A member can reserve a book only when every copy is out.
- Status moves through waiting → ready → fulfilled.
- When a copy returns, the oldest waiting member gets it.
- A ready reservation holds the copy for 2 days.
- Other members are blocked from taking a copy reserved for the next member.

11. JOINS and the Book Catalogue View

JOIN: Books + Authors

Matches books to authors using a

LEFT JOIN: Books + Categories

Keeps books even without a category

3-Table JOIN

Shows member, book and due date for

```
SELECT * FROM v_book_catalogue;
```

Saved query joining books, authors and categories
with an availability label.

12. Testing the System

- Borrowing with an unpaid fine is rejected.
- After the fine is paid, borrowing can proceed.
- A fully borrowed book is rejected when no copy is free.
- Returning a book can make a waiting reservation ready.
- A 16-day overdue example produces an 8,000 UGX fine.
- Audit records show actions including BORROW and PAYMENT.

13. DELIMITER, BEGIN and END

Used when defining procedures and triggers

DELIMITER

Temporarily changes the command

BEGIN

Starts a block of SQL statements.

END

Closes the block.

```
DELIMITER //  
CREATE PROCEDURE demo()  
BEGIN  
    SELECT 'Hello';  
    SELECT 'Library';  
END //  
DELIMITER ;
```

14. Current Limitations

Documented in the database guide

- `sp_borrow_book` is mentioned in the script header but is not included; borrowing uses an INSERT into loans.
- Ready reservations have a 2-day deadline but are not automatically cancelled.
- `sp_return_book` is silent for a wrong or already-returned loan ID.
- Fines are created when a late book is returned.
- There is no cancel-reservation procedure; cancelled status must be set manually.
- The audit log can become busy because borrowing and returning also create book-update records.

KEY TAKEAWAY

- The database stores library information in related tables.
- Constraints protect data integrity.
- Procedures perform reusable operations when called.
- Triggers enforce automatic business rules and record changes.
- Functions calculate values, while views simplify repeated queries.
- Together these features make the library system more consistent and easier to manage.

Source: Library Database — A Beginner's Guide